CAUTION: High voltages are present in this laboratory exercise! Do not make or modify any connections with the power on unless otherwise specified! If any danger occurs, immediately press the RED EMERGENCY OFF button on the Three-phase Power Supply Module. Complete this laboratory exercise as quickly as possible to avoid the rise in temperature under load condition. The machines are heavy, make sure to practice safety precautions when moving them from the cabinet to the table top.

Lab 1: DC Series Wound Machine

Activity 1.1: Motor mode – Torque-Speed Characteristic

1.1.1. Aim of activity

After completing this exercise, you should be able to demonstrate the torque-speed characteristic of DC series wound motor.

1.1.2. Equipment required

Table 1.1.1. List of equipment

Qty	Description	Cat. No.
1	DC Series Wound Machine	EM-3330-1E
1	Magnetic Powder Brake Unit	EM-3320-1C
1	Brake Controller	EM-3320-1N
1	DC Power Supply Module	EM-3310-1A
1	System Transformer	EM-3340-3B
1	Three-phase Power Supply Module	EM-3310-1E
1	Three-pole Current Limit Protection Switch Module	EM-3310-2A
1	Digital DCA Meter	EM-3310-3A
1	Digital DCV Meter	EM-3310-3B
1	Coupling	EM-3390-2A
1	Connecting Leads Set	EM-3390-3A
1	Safety Bridging Plugs Set	EM-3390-4A

1.1.3. Schematic

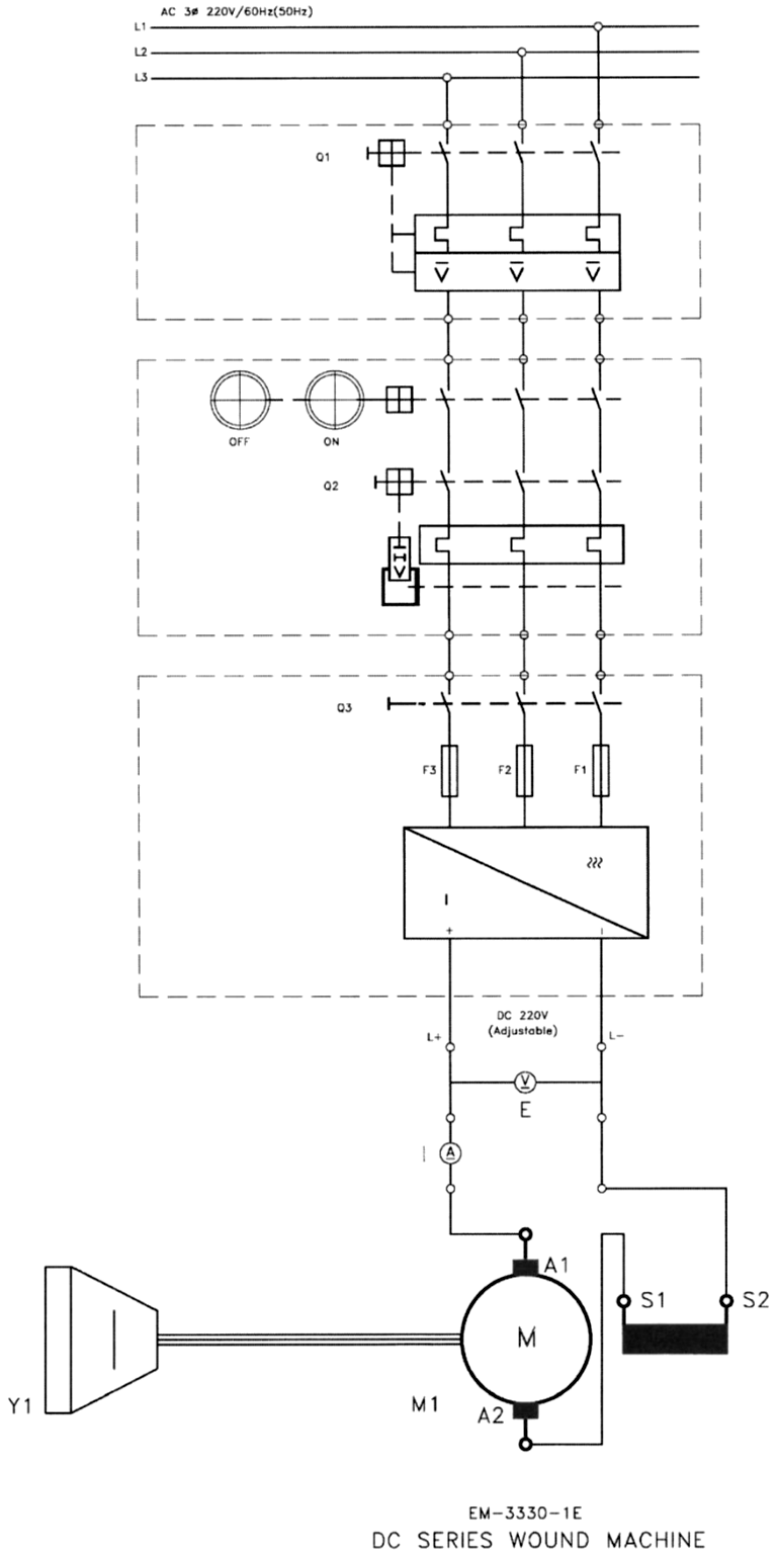


Figure 1.1.1. Circuit diagram for torque-speed characteristic test

1.1.4. Connection

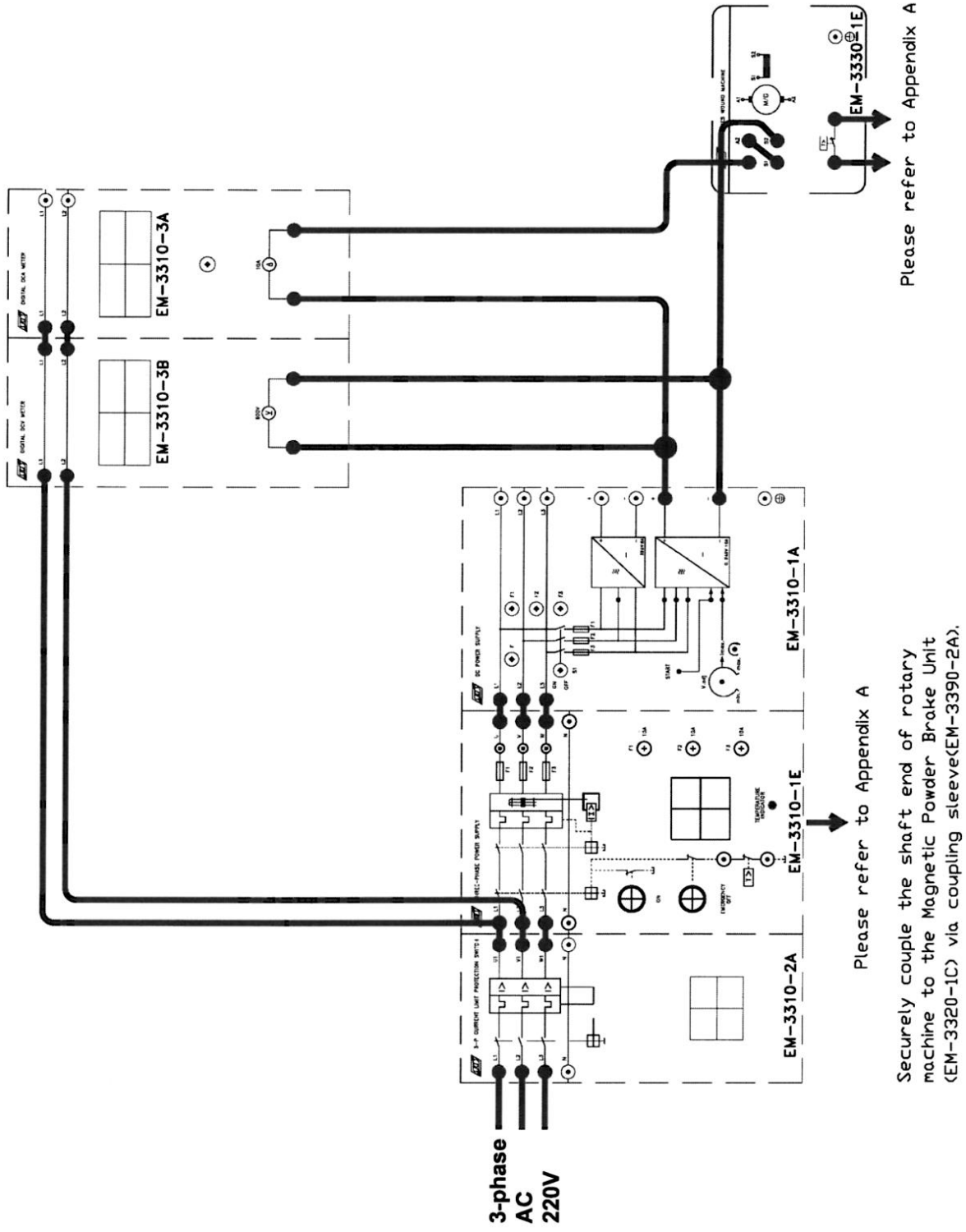


Figure 1.1.2. Connection diagram for torque-speed characteristic test

1.1.5. Experimental procedure

1. Place the DC Series Wound Machine, Magnetic Powder Brake Unit and Brake Controller on the Laboratory Table. Mechanically couple the DC Series Wound Machine to the Magnetic Powder Brake Unit using a Coupling. Securely lock the Machine Bases together using the delta screws. Electrically connect the Brake Controller to the Magnetic Powder Brake Unit using the supplied cable. **CAUTION: The series wound motor MUST NOT run with no mechanical load because the motor speed increases to a very high value which can damage the motor (motor runaway).**

2. Install the required Modules in the Experimental Frame. Construct the circuit in accordance with the circuit diagram and connection diagram in Figure 1.1.1 and 1.1.2, respectively. Have the instructor or lab technician to check on your completed circuit. **Note: The thermal switches of DC Series Wound Machine and Magnetic Powder Brake Unit MUST be connected together.**

3. On the DC Power Supply Module, set the V_{adj} knob to the min. position.

4. Sequentially turn on the main switches, Brake Controller, Magnetic Powder Brake Unit, System Transformer, 3-P Current Limit Protection Switch, Three-phase Power Supply, and DC Power Supply Modules.

5. Manipulate the Brake Controller to operate in Mode\Closed Loop\Constant Torque mode and press the up button to set the output torque T to 0.1 kg-m, then press Enter and you will hear a loud buzz sound, which is normal.

6. On the DC Power Supply Module, press the START button and slowly turn the V_{adj} knob to increase the motor voltage E up to the rated value of 220 Vdc. **Note: The motor speed (obtained from the Brake Controller) must not exceed 2,800 rpm. If the rotor is locked by a heavy brake torque, immediately press the red Emergency OFF button on the Three-phase Power Supply Module and terminate this exercise.**

7. Record the motor speed N (obtained from the Brake Controller), the motor current I (obtained from the Digital DCA Meter), and the motor voltage (obtained from the Digital DCV Meter)

in Table 1.1.2 in the Results and Discussions section. **Note: The torque value may fluctuate, wait awhile and take the most accurate values.**

8. Slowly return the V_{adj} knob on the DC Power Supply Module to the min. position. The Brake Controller will buzz again once the motor has stopped completely.

9. Release the braking by pressing the BACK button on the Brake Controller.

   DO NOT RELEASE THE BRAKE WHILE THE MOTOR IS STILL RUNNING. THE BRAKE HAS TO BE RELEASED FIRST TO SET NEW TORQUE VALUE, HENCE DO NOT INCREASE THE TORQUE WHILE THE MOTOR IS STILL RUNNING!

10. Repeat steps 5 through 9 for other torque settings listed in Table 1.1.2. **Note: The motor current I must not exceed 130% of the rated current (*1.65 A × 1.3 = 2.145 A*).**

11. Sequentially turn off the DC Power Supply, Three-phase Power Supply, 3-P Current Limit Protection Switch Modules, System Transformer, Magnetic Powder Brake Unit and Brake Controller.

1.1.6. Results and discussions

1. Fill in the values obtained from the previous section to Table 1.1.2 below:

Table 1.1.2. Measured values of I, E, and N

T (kg-m)	0.1	0.15	0.2	0.25
E (V)				
I (A)				
N (rpm)				

2. Using the results of Table 1.1.2, plot the graph of T vs. N and T vs. I and evaluate their relationship.

3. State the conclusion of this activity.

Activity 1.2: Motor mode – Speed Control

1.2.1. Aim of activity

After completing this exercise, you should be able to control the speed of DC series wound motor.

1.2.2. Equipment required

Table 1.2.1. List of equipment

Qty	Description	Cat. No.
1	DC Series Wound Machine	EM-3330-1E
1	Magnetic Powder Brake Unit	EM-3320-1C
1	Brake Controller	EM-3320-1N
1	DC Power Supply Module	EM-3310-1A
1	System Transformer	EM-3340-3B
1	Three-phase Power Supply Module	EM-3310-1E
1	Three-pole Current Limit Protection Switch Module	EM-3310-2A
1	DC Machine Starter	EM-3310-4A
2	Digital DCA Meter	EM-3310-3A
1	Digital DCV Meter	EM-3310-3B
1	Coupling	EM-3390-2A
1	Connecting Leads Set	EM-3390-3A
1	Safety Bridging Plugs Set	EM-3390-4A

1.2.3. Schematic

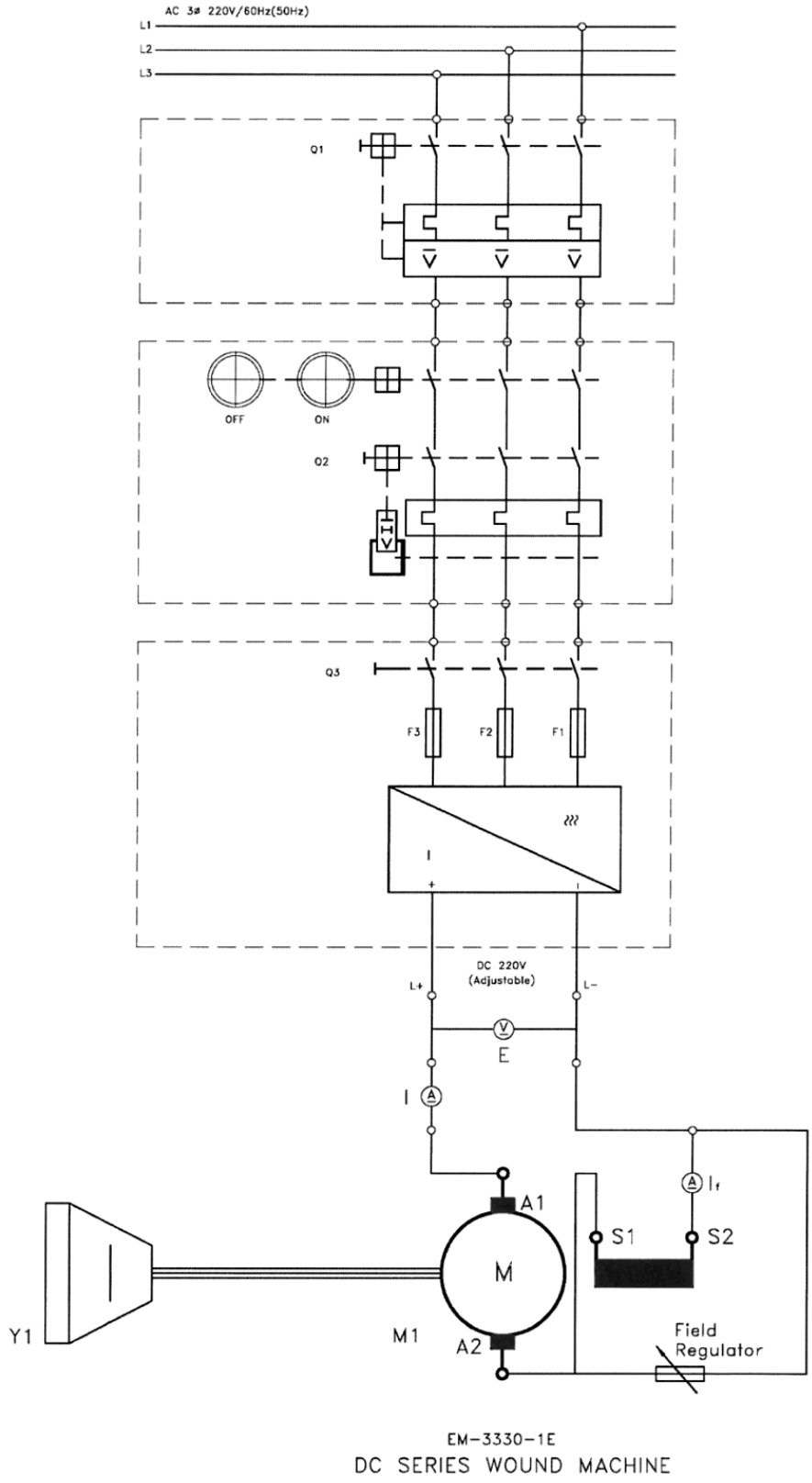


Figure 1.2.1. Circuit diagram for speed control

1.2.4. Connection

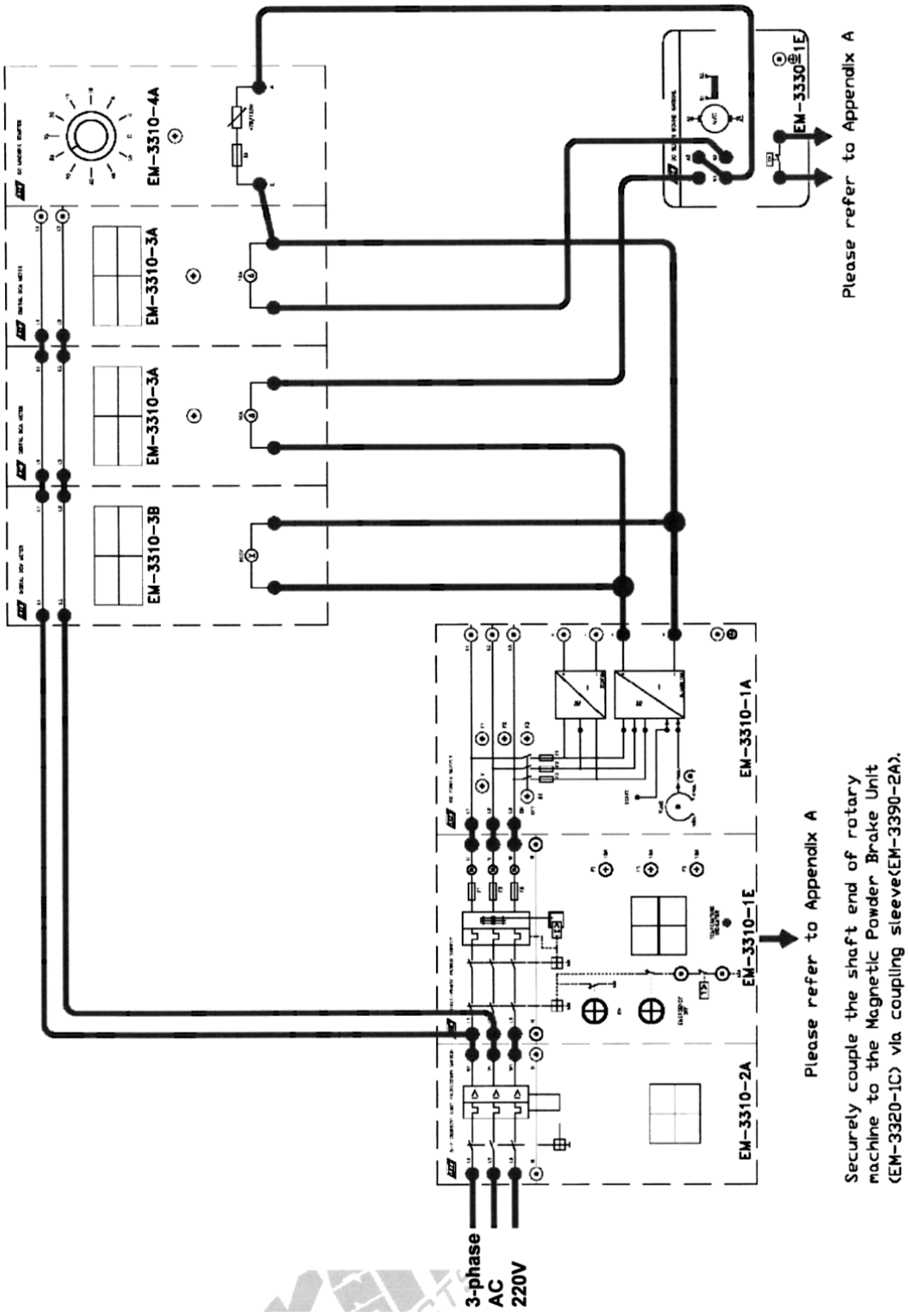


Figure 1.2.2. Connection diagram for speed control

1.2.5. Experimental procedure

1. Place the DC Series Wound Machine, Magnetic Powder Brake Unit and Brake Controller on the Laboratory Table. Mechanically couple the DC Series Wound Machine to the Magnetic Powder Brake Unit using a Coupling. Securely lock the Machine Bases together using the delta screws. Electrically connect the Brake Controller to the Magnetic Powder Brake Unit using the supplied cable. **CAUTION: The series wound motor MUST NOT run with no mechanical load because the motor speed increases to a very high value which can damage the motor (motor runaway).**

2. Install the required Modules in the Experimental Frame. Construct the circuit in accordance with the circuit diagram and the connection diagram in Figure 1.2.1 and 1.2.2, respectively. Have the instructor or lab technician to check your completed circuit. **Note: The thermal switches of DC Series Wound Machine and Magnetic Powder Brake Unit must be connected together.**

3. On the DC Power Supply Module, set the V_{adj} knob to the min. position.

4. On the DC Machine Starter, set the resistance knob to the max. (47 Ω) position.

5. Sequentially turn on the main switches, Brake Controller, Magnetic Powder Brake Unit, System Transformer, 3-P Current Limit Protection Switch, Three-phase Power Supply, and DC Power Supply Modules.

6. Manipulate the Brake Controller to operate in Mode\Closed Loop\Constant Torque mode and set the output torque T to 0.1 kg-m, then press Enter and you will hear a loud buzz sound, which is normal.

7. Press the START button on the DC Power Supply Module and slowly turn the V_{adj} knob to increase the motor voltage E up to the rated value of the motor, 220 Vdc. **Note: The motor speed (obtained from the Brake Controller) must not exceed 2,800 rpm. If the rotor is locked by a heavy brake torque, immediately press the red Emergency OFF button on the Three-phase Power Supply Module and terminate this exercise.**

8. Record the motor current I and the field current I_f (obtained from the Digital DCA Meters), the motor voltage E (obtained from the Digital DCV Meter), and the speed N (obtained from the Brake Controller) in Table 1.2.2. **Note: The motor current must not exceed 130% of the rated current (*1.65 A* × *1.3* = *2.145 A*).**

9. While the motor is running, gradually reduce the field resistances as listed in Table 1.2.2 in the Results and Discussions section and record down the values.

10. Slowly return the V_{adj} knob on the DC Power Supply Module to the min. position. The Brake Controller will buzz again once the motor has stopped completely.

11. Release the braking by pressing the BACK button on the Brake Controller. Turn the resistance knob on the DC Motor Field Regulator back to 47 Ω position.

    DO NOT RELEASE THE BRAKE BEFORE STOPPING THE MOTOR, TO AVOID MOTOR RUNAWAY.

12. Sequentially turn off the DC Power Supply, Three-phase Power Supply, 3-P Current Limit Protection Switch Modules, System Transformer, Magnetic Powder Brake Unit and Brake Controller.

1.2.6 Results and discussions

1. Fill in the values obtained from the previous section to Table 1.2.2 below:

Table 1.2.2. Measured values of I, I_f, E, and N

Ω knob	47	45	40	35	30	25	20	15	10	5	0
I (A)											
I_f (A)											
E (V)	220	220	220	220	220	220	220	220	220	220	220
N (rpm)											

2. Using the results of Table 1.2.2, plot the N vs. I and N vs. I_f curves, evaluate the relationship between the current and the speed of the DC motor with relevant theories and equations.

3. Distinguish between the curves obtained from this activity and textbook/literature. Explain and suggest an alternative method to conduct this experiment.

4. State the conclusion of this activity.

Activity 1.3: Generator mode – Load characteristic

1.3.1 Aim of activity

After completing this exercise, you should be able to demonstrate the operating characteristic of a DC series wound generator under load condition.

1.3.2. Equipment required

Table 1.3.1. List of equipment

Qty	Description	Cat. No.
1	DC Permanent-magnet Machine	EM-3330-1A
1	DC Series Wound Machine	EM-3330-1E
1	DC Power Supply Module	EM-3310-1A
1	System Transformer	EM-3340-3B
1	Three-phase Power Supply Module	EM-3310-1E
1	Three-pole Current Limit Protection Switch Module	EM-3310-2A
1	DC Generator Load Resistor	EM-3310-4H
2	Digital DCA Meter	EM-3310-3A
3	Digital DCV Meter	EM-3310-3B
1	Digital RPM Meter	EM-3310-3G
2	Coupling	EM-3390-2A
1	Connecting Leads Set	EM-3390-3A
1	Safety Bridging Plugs Set	EM-3390-4A

1.3.3. Schematic

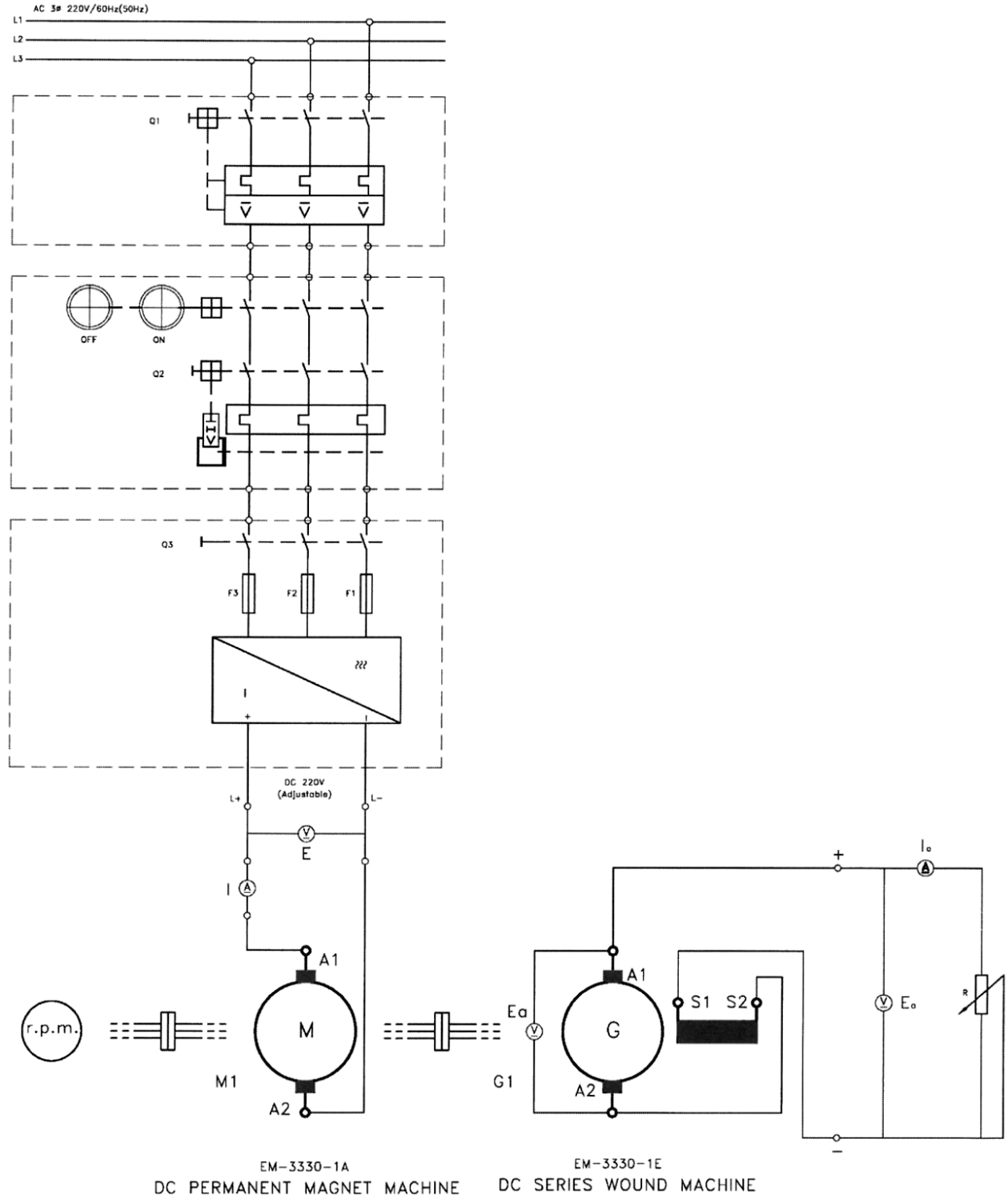


Figure 1.3.1. Circuit diagram for load characteristic test

1.3.4. Connection

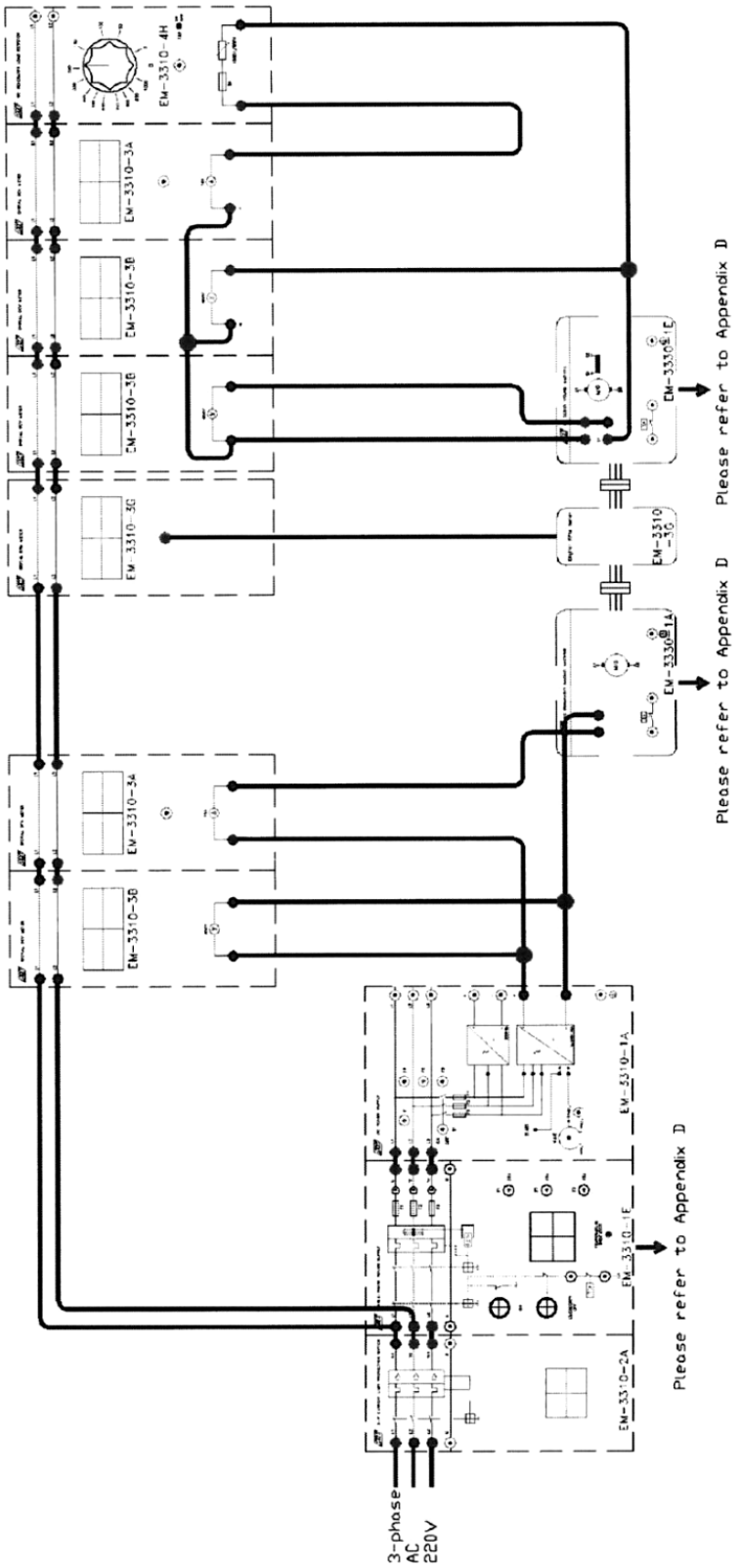


Figure 1.3.2. Connection diagram for load characteristic test

1.3.5. Experimental procedure

1. Place the DC Permanent-Magnet Machine (prime mover), DC Series Wound Machine, and Digital RPM Meter on the Laboratory Table. Mechanically couple the DC Permanent-Magnet Motor to the DC Series Wound Machine and the Digital RPM Meter using Couplings. Securely lock the Machine Bases together using delta screws. **Note: If the developed generator voltage is very small when the prime mover rotates at its rated speed, reverse the connecting leads to A1 and A2 terminals on the panel of the prime mover for building up generator voltage.**

2. Install the required Modules in the Experimental Frame. Construct the circuit in accordance the circuit diagram and the connection diagram in Figure 1.3.1 and 1.3.2, respectively. Have the instructor or lab technician to check on your completed circuit.

3. Set the V_{adj} knob on the DC Power Supply Module to the min. position. Set the resistance knob on the DC Generator Load Resistor to 1000 Ω position.

4. Sequentially turn on the main switches, System Transformer, 3-P Current Limit Protection Switch, Three-phase Power Supply, and DC Power Supply Modules.

5. Press the START button on the DC Power Supply Module. Then, on the DC Generator Load Resistor Module, set the FAN switch to the ON position.

6. On the DC Power Supply Module, slowly turn the V_{adj} knob to increase the motor voltage until the DC Permanent-Magnet Motor operates at the rated speed of 2,000 rpm. This speed must be maintained in the laboratory exercise. **Note: The motor current I must not exceed 130% of the rated value (*2.7 A × 1.3 = 3.51 A*), the generator output voltage E_o and current I_o must not exceed the rated value.**

7. Adjust the resistance knob on the DC Generator Load Resistor and set the generator output current I_o to 0.01 A. Record the generator output voltage E_o and the generator armature voltage E_a in Table 1.3.2 in the Results and Discussions section.

8. Repeat step 7 for other I_o settings listed in Table 1.3.2.

9. Slowly return the V_{adj} knob on the DC Power Supply Module to the min. position to stop the DC Permanent-Magnet Motor, and return the resistance knob on the DC Generator Load Resistor to 1000 Ω position.

10. Sequentially turn off the DC Power Supply, Three-phase Power Supply, 3-P Current Limit Protection Switch Modules, and System Transformer.

1.3.6 Results and Discussions

1. Fill in the values obtained from the previous section to Table 1.3.2 below and calculate the generator output power P_o:

Table 1.3.2. Measured values of E_o, E_a, and P_o

I_o (A)	0.01	0.2	0.4	0.6	0.8	1	1.2	1.4	1.6
E_o (V)									
E_a (V)									
P_o (W)									

2. Using the results from Table 1.3.2, plot the E_o vs. I_o, E_a vs. I_o, and P_o vs. I_o curves. Analyse the relationship between the current and voltage of the DC generator with relevant theories and equations.

3. State the conclusion of this activity.

Note: Consolidate all the lab activities 1.1, 1.2, and 1.3 into a single report.

APPENDIX 1

MARKING RUBRICS

Component Title	Lab report				Percentage (%)	100	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
Formatting, quality of figures and plots	All sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and mentioned in the main text. All figures are clear with important labels and comments. Title and the axis of all plots are well labelled, clear, and easy to read.	Most sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and most are mentioned in the main text. Most figures are clear with labels and comments. Title and the axis of most plots are well labelled and clear.	Some sections are inconsistent in terms of format and style. Most captions for tables and figures are given and mentioned in the main text. Some figures and components in the circuit diagram are not clear and missing important labels and comments. Some title and axis of plots are not well labelled and clear.	Most sections are inconsistent in terms of format and style. Incomplete captions for tables and figures and most are not mentioned in the main text. Most figures and components in the circuit diagram are not clear and missing important labels and comments. Most title and axis of plots are not well labelled and clear.	Messy report and difficult to read with inconsistent format and style. No captions for tables and figures provided. Blurry figures and plots with missing labels provided. Figures and plots are difficult to read and understand.	15	
Introduction and Methodology	Excellently written, concise introduction and methodology with important information that are relevant to the topic.	Well written, concise introduction and methodology with important information that are relevant to the topic.	Averagely written, concise introduction and methodology, consists some irrelevant information.	Averagely written, lengthy introduction and methodology, consists many irrelevant information.	Poorly written, lengthy introduction and methodology, consists many irrelevant information.	15	
Results and discussions	Completed all experiments and data collection. All measured data, calculations, and graphs/plots are correct. Able to evaluate the results with prudent judgment. Analysis is done with excellent description of the relationship involved and explanation with relevant theories.	Completed all experiments and data collection. Most measured data, calculations, and graphs/plots are correct. Results are compared and analysed with good description of the relationship involved and with relevant theories.	Completed most experiments and data collection. Some measured data, calculations, and graphs/plots are incorrect. Results analysis are done with adequate description of the relationship involved.	Completed most experiments, but most measured data, calculations, and graphs/plots are incorrect. Only restatement of the results without commenting on the expected key points.	Experiments not completed. Most measured data, calculations, and graphs/plots are incorrect. Discussion shows little understanding of what the experiment is all about.	50	
Conclusion	The aim of the experiment has been achieved, the key features of the methods used and most important results are summarized.	The aim of the experiment has been achieved, most key features of the methods used and results are summarized.	A conclusion is drawn, but only supported by some experimental evidence.	A conclusion is drawn, but not supported by any experimental evidence.	No sensible conclusion. No comparisons and evidence.	20	
					TOTAL	**100**	

Note to students: Please attach this appendix together with the submission of coursework.